

Lux Sensor is a high accuracy ambient light digital 16-bit resolution sensor in a miniature transparent 6.8 mm x 3.0 mm x 2.5 mm package. It includes a highly sensitive photo diode, a low noise amplifier, a 16-bit A/D converter and supports an easy to use I2C bus communication interface.

Key Features

- Integrated solution combining proximity and ambient light sensing in one package
- High sensitivity for accurate detection of objects and light levels
- Wide detection range suitable for consumer and industrial applications
- I²C interface for simple integration with microcontrollers
- Low power consumption optimized for portable and battery-powered devices
- Interrupt functions for proximity and ambient light thresholds, reducing MCU load
- Compact package enabling space-saving designs



VCNL4040 Sensor Board

Technical Specifications

Parameter	Value
Proximity Sensor Range	Up to 200 mm with sunlight immunity
Ambient Light Sensor Range	0.0125 lux to 20,000 lux (16-bit resolution)
Supply Voltage	2.5 V to 3.6 V
Communication Interface	I ² C, up to 400 kHz
Package Type	8-pin lead-free, compact surface-mount package
Operating Temperature	−40 °C to +85 °C
Current Consumption	~80 μA (ALS), ~200 μA (Proximity), <1 μA in shutdown mode

Applications

- **Smartphones & Tablets** — Auto screen brightness control, proximity sensing during calls.
- **Wearables** — Gesture recognition and power-saving display control. And many more.